Select exactly one answer per question.

1. Which quantity is decisive for the thermal load of a current-carrying conductor?

A) the electric field strength

B) the electrical power

C) the electrical current density

D) the electrical resistance

Solution: C

2. An ideal ohmic resistor is characterized by the fact that

A) current and voltage are proportional to each other.

B) a phase shift occurs between current and voltage.

C) the resistance value is time-dependent.

D) electrical energy is periodically stored and released.

Solution: A

3. Which statement describes the I–V characteristic of an ideal current source?

A) The voltage is proportional to the current.

B) The current is independent of the terminal voltage.

C) The characteristic passes through the origin.

D) The characteristic is frequency-dependent.

Solution: B

4. Which statement about electrical power and energy is correct?

A) Power and energy are independent of each other.

B) Energy is the time derivative of power.

C) With constant power, energy increases linearly with time.

D) Power and energy differ only by a constant factor.

Solution: C

5. Kirchhoff's voltage law is a consequence of:

A) conservation of charge

B) conservation of momentum

C) conservation of temperature

D) conservation of energy

Solution: D

6. A Y–Δ transformation is permissible in a DC network if

A) all resistors have the same value.

B) the network is supplied by only one voltage source.

C) no power is dissipated in the network.

D) the current–voltage relationships at the external terminals remain unchanged.

Solution: D

7. Ideally, a voltmeter has:

A) a small internal resistance

B) a large internal resistance

C) no internal resistance

D) a frequency-dependent resistance

Solution: B

8. Which statement correctly describes the efficiency of an electrical system?

A) It is the ratio of output power to input power.

B) It is the ratio of reactive power to active power.

C) It is independent of the electrical losses in the system.

D) In AC operation, it can be greater than one.

Solution: A

9. Which statement about electric and magnetic field lines is correct?

A) Electric field lines are always closed.

B) Magnetic field lines are always closed.

C) Magnetic field lines end at magnetic charges.

D) Electric field lines exist only inside conductors.

Solution: B

10. Which of the following quantities is a directly measurable physical field in macroscopic electrodynamics?

A) electric displacement $\mathbf{D}$

B) magnetic field strength $\mathbf{H}$

C) magnetic flux density $\mathbf{B}$

D) electric polarization $\mathbf{P}$

Solution: C

11. A time-varying magnetic field generates, according to Faraday's law:

A) an induced voltage

B) a constant current

C) heat

D) a mechanical force

Solution: A

12. Why can the voltage across an ideal capacitor not change abruptly?

A) Because the capacitor blocks direct current, rapid voltage changes are not possible.

B) An ideal capacitor fundamentally prevents any change in the stored energy.

C) An abrupt change in voltage would require an infinitely large current.

D) The voltage across a capacitor is limited from above by the capacitance.

Solution: C

13. Under constant DC voltage, an ideal inductor in the steady state behaves like:

A) an open circuit

B) a current source

C) a voltage source

D) a short circuit

Solution: D

14. A capacitor is discharged through an ohmic resistor. Which statement correctly describes the discharge process?

A) The voltage across the capacitor decreases exponentially, while the current increases exponentially.

B) The voltage across the capacitor increases exponentially, while the current decreases exponentially.

C) The voltage across the capacitor decreases exponentially, and the current also decreases exponentially.

D) The voltage across the capacitor increases exponentially, and the current also increases exponentially.

Solution: C

15. Why is the RMS value typically specified for mains voltages?

A) Because it is smaller than the peak value.

B) Because it describes the power effect of the voltage in ohmic loads.

C) Because it is independent of frequency.

D) Because it completely describes the time dependence of the voltage.

Solution: B

16. For an ideal inductor in an AC circuit, the following applies:

A) The current leads the voltage.

B) The current lags behind the voltage.

C) Current and voltage are phase-shifted by 45° relative to each other.

D) Current and voltage are in phase.

Solution: B

17. What physical cause leads to the current leading the voltage in a capacitor?

A) The voltage causes an immediate current flow through the dielectric.

B) The voltage is delayed by ohmic losses in the capacitor.

C) Magnetic effects determine the behavior of the capacitor.

D) The current charges the capacitor, causing the voltage to build up with a delay.

Solution: D

18. What physical cause leads to the occurrence of reactive power in an AC circuit?

A) Electrical energy is periodically stored in fields and released again.

B) Electrical energy is periodically stored in heat and released again.

C) Electrical energy is periodically returned to the source and stored.

D) Electrical energy is periodically transported in the conductor and delayed.

Solution: A

19. What filter characteristic is exhibited by a parallel connection of an inductor and a capacitor when it is connected in series with a load?

A) Low-pass

B) High-pass

C) Band-pass

D) Band-stop

Solution: D

20. A symmetrical three-phase system:

A) generates a rotating electric field

B) requires a neutral conductor only with an unbalanced load

C) operates with direct current

D) has no voltage between the line conductors

Solution: B